Instructions:

1. This question paper contains a total of **1** page (**1** side of paper).
2. Write your name, roll number, department, and section on every side of every sheet of this booklet
3. Write final answers neatly with a blue/black pen in the given boxes.
4. **Answers written outside the box will NOT be graded.**

Total **10 Marks**

Q. 1: Write the output of the following program in the appropriate box and answer the question. (2+2+2 = 6 Marks)

```
1   #include<stdio.h>
2   void func1(int *arr){
3     for(int i=1; i<=3;i++)
4       (i-1)[arr] = (i%3);
5   }
6   void func2 (int a[1] , int arr){
7     a[0] = arr + 1;
8   }
9  int main(){
10   int arr[4] = {0,0,0,0};
11   func1(arr+1);
12   for(int i=0; i<4; i++)
13          printf("%d ", i[arr] );
14   printf("\n");
15   func2(arr, *arr);
16   for(int i=0; i<4; i++)
17          printf("%d ", i[arr] );
18   return 0;
19 }
```

arr(0,0,0)

arr[0] = 1%3 = 1
@ 1+0 = 1%3
a0=0.
a1=1
a2=2
a3=0

arr[0] = arr[1] 1
arr[1] = ~~arr~~ 2.
arr[2] = 0+1
arr[3]

arr arr[1] arr[2] arr[3]

Output
0120
~~1200~~ 1111.

Will the answer change if we write func2 as:

```
void func2 (int a[1] , int arr){
  int temp = arr + 1;
  a = &temp;
}
```

Explain your answer

Yes the answer would change as we are assigning the address to a, while in the code written on left we are assigning the value of arr[0]+1 to a.

Q. 2: Write the output of the following sequence of instructions **long arr2[100]; int arr1[100], a1=(&arr2[50]), a2=(&arr2[10]), b1=(&arr1[50]), b2=(&arr1[10]); printf("%d %d", a1-a2 , b1-b2);** Explain your answer. (2+2 Marks)

320 160

Since $a_1 - a_2$ is the difference between the addresses of arr2[50] ~~and~~ arr2[10] and since arr2 lies in ~~double~~ long therefore diff between consecutive addresses is equal to 8 bytes. And the difference between the consecutive addresses storing integer values is 4.

10
1
2
3
4
5
6
7
8
9
10

40x4=160
8x40=320